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Part 2 Report

Output Screenshots:

```
barch@ShirasPro:~$ python3 DNS_client_shira_alysa.py google.com
Querying 198.41.0.4 for google.com
-----
NS : l.gtld-servers.net
NS : j.gtld-servers.net
NS : h.gtld-servers.net
NS : d.gtld-servers.net
NS : b.gtld-servers.net
NS : f.gtld-servers.net
NS : k.gtld-servers.net
NS : m.gtld-servers.net
NS : i.gtld-servers.net
NS : g.gtld-servers.net
NS : a.gtld-servers.net
NS : c.gtld-servers.net
NS : e.gtld-servers.net
A : 192.41.162.30
AAAA : 2001:500:d937::30
A : 192.48.79.30
AAAA : 2001:502:7094::30
A : 192.54.112.30
AAAA : 2001:502:8cc::30
A : 192.31.80.30
AAAA : 2001:500:856e::30
A : 192.33.14.30
AAAA : 2001:503:231d::2:30
A : 192.35.51.30
RTT: 36.47 ms
-----
Querying 192.41.162.30 for google.com
-----
NS : ns2.google.com
NS : ns1.google.com
NS : ns3.google.com
NS : ns4.google.com
AAAA : 2001:4860:4802:34::a
A : 216.239.34.10
AAAA : 2001:4860:4802:32::a
A : 216.239.32.10
AAAA : 2001:4860:4802:36::a
A : 216.239.36.10
AAAA : 2001:4860:4802:38::a
A : 216.239.38.10
RTT: 33.40 ms
-----
Querying 216.239.34.10 for google.com
-----
A : 142.250.189.174
RTT: 39.93 ms
-----
Making HTTP request to 142.250.189.174
-----
HTTP/1.1 301 Moved Permanently
RTT: 49.93 ms
-----

barch@ShirasPro:~$ python3 DNS_client_shira_alysa.py wikipedia.org
Querying 198.41.0.4 for wikipedia.org
-----
NS : a2.org.afilias-nst.info
NS : b2.org.afilias-nst.org
NS : d0.org.afilias-nst.org
NS : a0.org.afilias-nst.info
NS : b0.org.afilias-nst.org
NS : c0.org.afilias-nst.info
A : 199.249.112.1
AAAA : 2001:500:40::1
A : 199.249.120.1
AAAA : 2001:500:48::1
A : 199.19.57.1
AAAA : 2001:500:f::1
A : 199.19.56.1
AAAA : 2001:500:e::1
A : 199.19.54.1
AAAA : 2001:500:c::1
A : 199.19.53.1
AAAA : 2001:500:b::1
RTT: 26.61 ms
-----
Querying 199.249.112.1 for wikipedia.org
-----
NS : ns0.wikimedia.org
NS : ns1.wikimedia.org
NS : ns2.wikimedia.org
A : 208.80.154.238
AAAA : 2620:0:861:53::1
A : 208.80.153.231
AAAA : 2620:0:860:53::1
A : 198.35.27.27
AAAA : 2a02:ec80:53::1
RTT: 93.13 ms
-----
Querying 208.80.154.238 for wikipedia.org
-----
A : 198.35.26.96
RTT: 84.74 ms
-----
Making HTTP request to 198.35.26.96
-----
HTTP/1.1 301 Moved Permanently
RTT: 36.66 ms
-----
```

1. The code for the DNS program was built manually using raw bytes and the struct.pack, as per the assignment regulations. The DNS header in particular contains a randomly generated transaction ID which matches requests and responses, flags set to 0x0000 to indicate a query and one question QDCOUNT set to one, with zero answer authority and additional records. The question section includes the domain name using DNS label format, a terminating byte indicating the end of the domain name, a query type set to A, like an IPv4 address, and query class set to IN, as in internet.
2. The DNS response packet is parsed manually using raw bytes. The DNS header was unpacked to determine the number of answer authority and additional records in it. Domain names are then subsequently decoded using a custom name-parsing function. Each resource record is parsed to extract record type, like A, NS, CNAME, AAAA,

etcetera, and the associated value of each record resource, like IP address or domain name. Records are stored separately based on if they appeared in the answer, authority, or additional section. The resolver starts by sending a DNS query to a IANA root server. If no answer is received, it either follows referrals from the root server to the appropriate TLD server, and then the TLD server refers the resolver to the authoritative name server. If IP addresses in the additional section are present they are used directly, but if a domain name has no such addresses the program resolves the nameserver host name by recursively performing DNS resolution. The process continues until an A record, i.e. a final IP address is obtained.

3. After resolving the final IP address, a TCP connection is created using packets. Afterwards, a raw HTTP GET request is made and sent to port 80. The HTTP status line, HTTP/1.1 301 Moved permanently is subsequently extracted from the response, alongside the RTT for the HTTP request, which is measured and reported in the following line.
4. External resources used beyond lecture notes and reading provided by the professor is ChatGPT, used for debugging purposes and guidance regarding errors encountered in the code, as per ACM regulations
5. RTTs from the DNS server:

	Root DNS server RTT	TLD DNS server RTT	Authoritative DNS server RTT
google.com	36.47 ms	33.40 ms	39.93 ms
wikipedia.org	26.61 ms	93.13 ms	84.74 ms

6. RTT from the requested domain:

	Requested domain RTT
google.com	49.93 ms
wikipedia.org	36.66 ms